

→ Regression
Classification

$$\left(\frac{80}{100}\right)$$

$$R^2 = \frac{\text{Regressed SS}}{\text{Total SS}} = 1 - \frac{\text{SSE}}{\text{TSS}}$$

Squared Error

How much variance explanation is done? $\Rightarrow R^2$

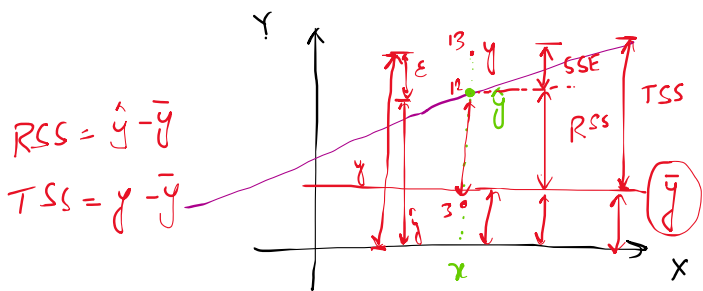
Metrics

Quality = Error
 R^2 , Adjusted R^2
↓
of rows
of columns

$$y = \hat{y} + \epsilon$$

$$TSS = RSS + SSE$$

(Regression)



$$y = \hat{y} + \epsilon$$

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

$$\beta = (X^T X)^{-1} X^T Y$$

$$y = \hat{y} + \epsilon$$

$$\hat{y} + TSS = \hat{y} + RSS + SSE$$

TSS - Total Sum of Squares

RSS - Regressed " " "

SSE - Sum of Squared Errors

$$TSS = RSS + SSE$$

$$\Rightarrow \because SSE = 0 \Rightarrow TSS = RSS$$

Algo → Train
↓
model
Train Test

$$\text{Coef. of determination } (R^2) = \frac{RSS}{TSS}$$

$$0 \leq R^2 \leq 1$$

of rows

$$y = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x-\mu}{\sigma}\right)^2}$$

of rows
of columns

Curse of dimensionality

$R^2 \leftarrow \text{train}$
 $\text{df.shape[0]} \rightarrow \text{rows} = n$
 $\text{df.shape[1]} \rightarrow \text{cols} = k$
 $\rightarrow \text{Adjusted } R^2 = 1 - \frac{(1-R^2)(n-1)}{(n-k-1)}$

Train $R^2 \leftarrow > 2\%$
 Test $R^2 \leftarrow X$

$R^2 \cong \text{Adj } R^2$
 $\left\{ \begin{array}{l} 0.7219 \\ 0.7257 \end{array} \right\}$
 $\frac{R}{P_y} \rightarrow \text{stats model.}$
 $\rightarrow \text{ sklearn (regularization)}$

$n \rightarrow \# \text{ of rows}$
 $k \rightarrow \# \text{ of columns}$

$\text{Train } R^2 \leftrightarrow \text{Adj } R^2$
 $\text{Test } R^2 \leftrightarrow \text{Adj } R^2$

Error Metrics

- 1) Root Mean Square Error
- 2) " " " "
- 3) " Absolute " "

$\left[\frac{\sum (y - \hat{y})^2}{n} \right]^{1/2}$

$\frac{\sum (y - \hat{y})^2}{n}$

$\frac{\sum |y - \hat{y}|}{n}$

$\rightarrow \left\{ \begin{array}{l} 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100 \end{array} \right\}$

$R^2 = 0.84$

$\text{RMSE} = 200,000$

$\log(y)$
 $\rightarrow e^{\hat{y}} =$

Scale of y
 $(y) \uparrow$
 (β)

Classification

0 - neg
1 - pos

Actual	Predicted	
N	0	True Negative (TN) ✓
0	0	" " (TN) ✓
0	0	" " (TN) ✓
0	1	False Positive (FP)
P	1	True Positive (TP) ✓
1	0	False Negative (FN) ✓
1	0	" " (FN) ✓

$$P = TP + FN$$

$$N = TN + FP$$

1-1 → TP ↑ Sn ↑
0-1 → FP ↑ TN ↓ Sp ↓

$$Accuracy(Ac) = \frac{TP + TN}{P + N} = \frac{1 + 3}{7} = \frac{4}{7}$$

$$\text{Sensitivity (Sn)} = \text{Recall (Re)} = \text{true positive rate (tpr)} = \frac{TP}{P} = \frac{TP}{TP + FN} = \frac{1}{3}$$

$$\text{Specificity (Sp)} = \text{true negative rate (tnr)} = \frac{TN}{N} = \frac{TN}{TN + FP} = \frac{3}{4}$$

$$\text{Precision (Pr)} = \frac{TP}{TP + FP} = \frac{1}{2}$$

$$F1 = 2 * \frac{Re * Pr}{Re + Pr}$$

Balance in Re & Pr

{ Sn - Sp } ←

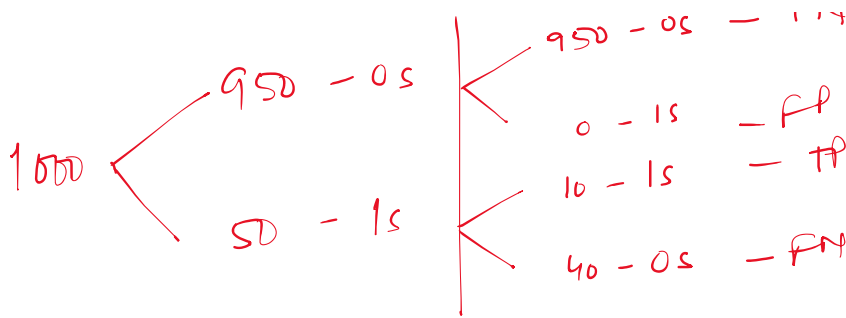
{ Re - Pr } ←

Acc = 96% Is it a good model??

1000	950 - 0s	950 - 0s - TN
		0 - 1s - FP
	50 - 1s	10 - 1s - TP
		40 - 0s - FN

$$Ac = \frac{10 + 950}{1000} = 96\%$$

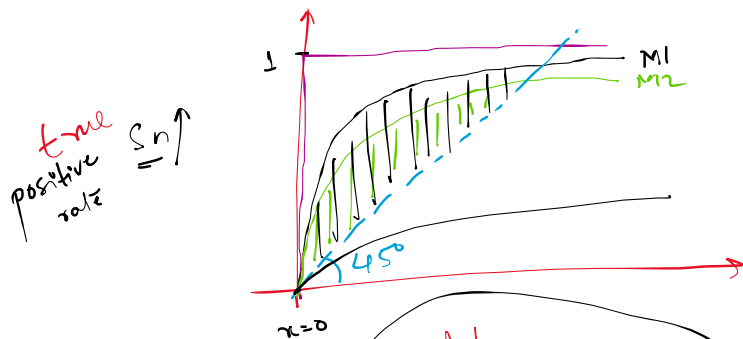
$$Re = \frac{10}{50} = 20\%$$



$$AC = \frac{10}{1000} = 1\%$$

$$Re = \frac{10}{50} = 20\%$$

Receiver Operator Characteristics (ROC) Curve

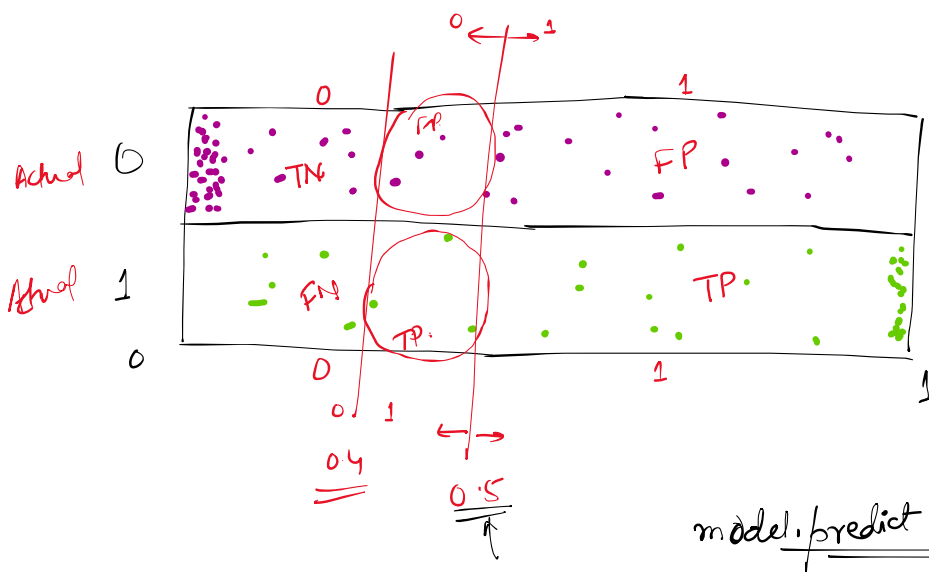


Area under curve
 $AUC(M1) > AUC(M2)$
 M1 is a better model.

$$100\% \left\{ \begin{aligned} S_n &= \frac{TP}{TP+FP} = 1 \\ S_p &= \frac{TN}{TN+FP} = 1 \end{aligned} \right.$$

$$\begin{aligned} 1 - \frac{TN}{N} \\ = 1 - \frac{TN}{TN+FP} \\ = \frac{TN+FP-TN}{N} \\ = \frac{FP}{N} \end{aligned}$$

Optimal Thresholds.

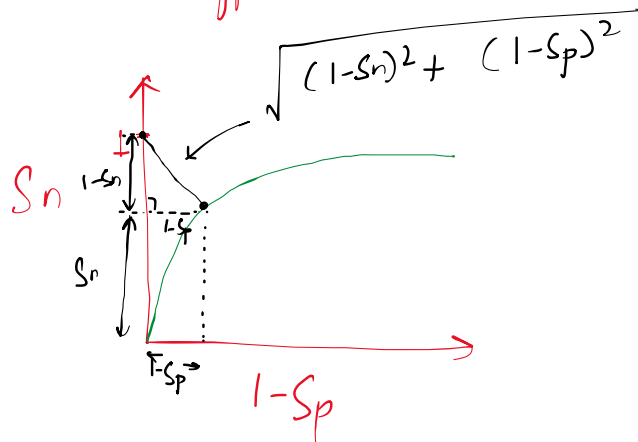


$$Re = \frac{TP}{N}$$

$Re \uparrow$

Value Calc

- ✓ 1. Distance cutoff
- ✓ 2. KS Cutoff (Kolmogorov - Smirnov)
3. Re-Pr / F1 cutoff



prob where d is minimum is dist cutoff.

KS Cutoff

Principle: O_s & I_s should have $\max^m$ separation.

1000 $\left\{ \begin{array}{l} 700 - O_s \\ 300 - I_s \end{array} \right.$

0 0.20 ✓
 0 0.90 ✓
 1 0.01 ✓
 1 0.99 ✓
 0 0.02 ✓
 1 0.98 ✓

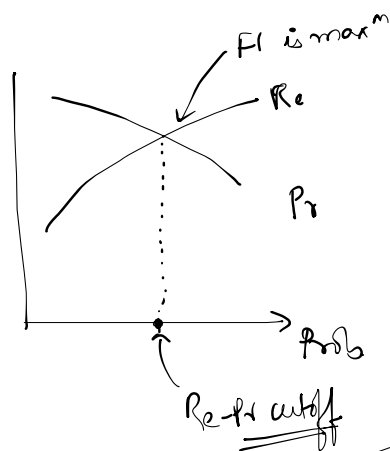
- ✓ 1. Predict probability
- ✓ 2. Sort in a descending order of prob
- ✓ 3. Form Bins

O_s $\downarrow$ I_s $\downarrow$ L_0

Bin#	# O_s	# I_s	% O_s	% I_s	Cum% I_s	Cum% O_s	Diff
1	0	100	0/700 = 0	100/300 = 0.33	0.33	0.00	0.33
2	10	90	10/700 = 0.01	90/300 = 0.30	0.63	0.01	0.62
3	30	70	30/700 = 0.02	70/300 = 0.18	0.81	0.03	0.78
4	80	20	80/700 = 0.10	20/300 = 0.12	0.93	0.13	0.80
5	90	10					
...	...	...					
...	...	...					
...	...	...					
10	100	0			1.00	1.00	0.00

→ prob where $\max (tpr - fpr)$ is $\max^m$ is KS cutoff

Re - Pr



prob where F1 is max^m
is F1 cutoff

f1-cutoff = 0.52

SHAP

row #
1
:
:
:
:
:
:
1000

prob	label
0.91	1
0.78	1
0.78	1
0.30	0
	:

crosstable

Actual

Confusion Matrix

	Predicted	
	0	1
0	TN	FP
1	FN	TP

Recall

Value calculation

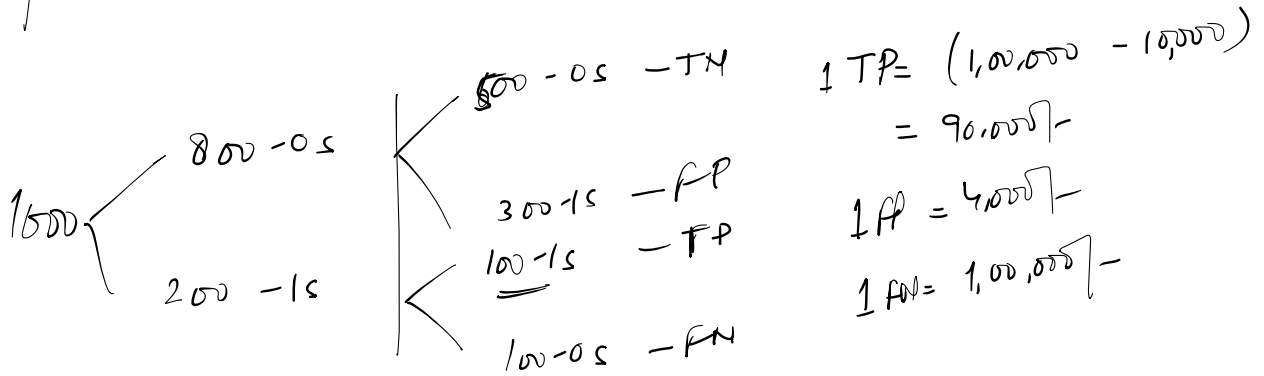
m/c failure

Replacement - 1,00,000/-
Repair - 10,000/-
Inspection - 4,000/-

0 — 0 → TN → No Loss / Profit
1 — 1 → TP → Repaired / Repair Cost
1 — 0 → FN → Replacement Cost
0 — 1 → FP → Inspection Cost

0.5 - TN 1 TP = (1,00,000 - 10,000)

unspec...



xs are 50% recd

Profit $100 \times 90,000 = 90,00,000$ —

Loss $300 \times 4000 + 100 \times 1,00,000$
 $1,00,00,000$
 $12,00,000$

 $1,12,00,000$ — Loss

Regularization

$$\hat{y} = \beta_0 + \beta_1 x_1 + \dots + \beta_n x_n$$

$$y \uparrow = \beta_1 x_1$$

$\beta_1 = 500$
 $\beta_1 = 1.2$

$x_1 \uparrow 1 \text{ unit}$
 $x_1 \uparrow 1 \text{ unit}$

$y = ?$
 $y = ?$



$$\epsilon = (y - \hat{y})^2$$

$$= \left(y - (\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n) \right)^2$$

$$= \left(\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} - \begin{bmatrix} 1 & x_{11} & x_{12} & x_{13} \\ 1 & x_{21} & x_{22} & x_{23} \\ 1 & x_{31} & x_{32} & x_{33} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix} \right)^2$$

$$-X^T Y + X^T X \beta + \lambda I \beta = 0$$

$$(X^T X + \lambda I) \beta = X^T Y$$

$$\beta \downarrow = \underline{(X^T X + \lambda I)^{-1}} \underline{X^T Y}$$

$$\begin{array}{l} \lambda \uparrow \quad \beta \downarrow \\ \lambda \downarrow \quad \beta \uparrow \end{array}$$

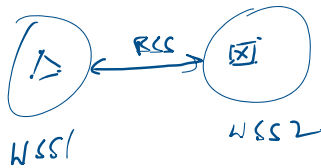
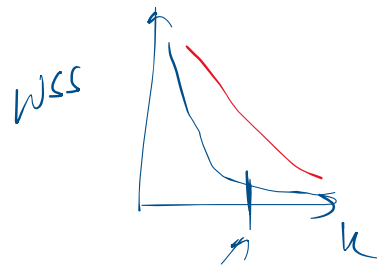
1) Scale data

2) label encoding

3) determine optimal k

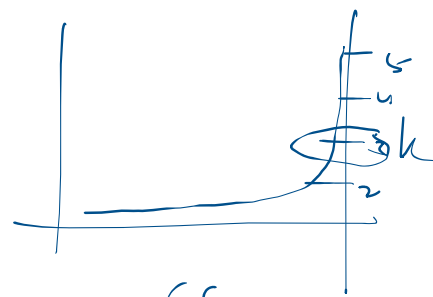
① Elbow method

within sum of squares



Within SS
Between SS

② Calinski method
calinski score $\propto \frac{BSS \uparrow}{WSS \downarrow}$



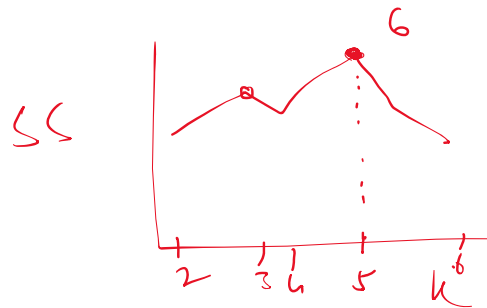


$$\textcircled{3} \text{ Silhouette score} = \frac{b-a}{\max(a,b)} = \frac{BSS-WSS}{\max(WSS, BSS)}$$

$$\cdot \underline{WSS=0} \quad SS = \frac{BSS}{\max(0, BSS)} = \frac{BSS}{BSS} = 1$$

$$\cdot BSS=0 \quad SS = \frac{-WSS}{\max(WSS, 0)} = -\frac{WSS}{WSS} = -1$$

$$-1 \leq SS \leq +1$$



④ Build clusters

⑤ Do cluster profiling

		groupby (cluster number)				mean of all cols
	for	x_2	x_3	x_4	x_5	# of rows
C0	mean	—	—	—	—	—
C1	—	—	—	—	—	—
C2	—	—	—	—	—	—
C3	—	—	—	—	—	—

(6) Box plot / Bar plot



(7) Insights & Recommendations